

*Please attach this cover sheet to your completed homework*

**Homework 1 (Ch. 7)**

**ECE 311 – Hang Gao**

Name: Gianni Avilan

| Problem | Course outcome | Grade |
|---------|----------------|-------|
| 1       | 1.c.           | /20   |
| 2       | 1.c.           | /20   |
| 3       | 1.c.           | /20   |
| 4       | 1.c.           | /20   |
| 5       | 1.c.           | /10   |
| 6       | 1.c.           | /10   |
| TOTAL:  |                | /100  |

ABET Assessed Outcomes

1.c. Use appropriate models to formulate solutions related to electric circuit problems.

## Homework #1

Due date: Sunday, Feb. 1<sup>st</sup> by 11:59 PM (Electronically – Canvas)

**1. (20 points)** The voltage and current of an electric load can be expressed by the following equations:

$$v = 340 \sin(628.318t + 0.5236) \text{ V}$$

$$i = 100 \sin(628.318t + 0.87266) \text{ V}$$

Note that the angles are in radians.

Calculate the following:

- The rms voltage
- The frequency of the current
- The phase shift between current and voltage in degrees
- The average voltage
- The load impedance

**2. (20 points)** An electric load consists of a  $4 \Omega$  resistance, a  $6 \Omega$  inductive reactance, and an  $8 \Omega$  capacitive reactance connected in series. The total impedance of the load is connected across a voltage source of 120V:

- Compute the power factor of the load
- Compute the source current
- Compute the real power of the circuit
- Compute the reactive power of the circuit

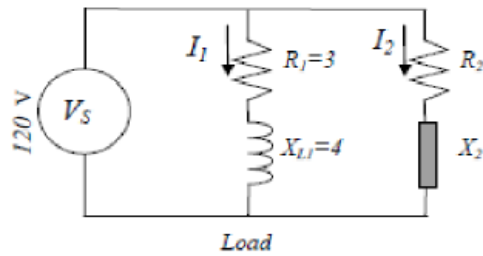
**3. (20 points)** A voltage source  $\bar{V} = 120 \angle 30^\circ$  is connected across a circuit consisting of  $R = 3.5 \Omega$  in series with  $L = 0.0833 \text{ H}$ . If the circuit is energized in the united states (the power supply frequency is 60Hz), compute the following:

- The load current
- The load power factor
- The real power consumed by the load
- The reactive power consumed by the load

**4. (20 points)** A voltage source  $\bar{V} = 120 \angle 30^\circ$  is connected to a circuit consisting of  $R = 3.5 \Omega$  in series with  $L = 0.0833 \text{ H}$ . If the circuit is energized in Europe (the power supply frequency is 50Hz), compute the following:

- The load current
- The load power factor
- The real power consumed by the load
- The reactive power consumed by the load

**5. (10 points)** An inductive load consists of  $R_1 = 3 \Omega$  in series with  $X_{L1} = 4 \Omega$ . The load is connected in parallel with another load of unknown impedance ( $R_2 + jX_2$ ). The voltage source of the system is 120V. The total real and reactive powers delivered by the source are 3 kW and 2 kVAR, respectively. Compute the impedance of the unknown load.



- 6.** (10 points) An inductive load consists of  $1\ \Omega$  resistance and  $1.0\text{mH}$  inductance connected in series. The load is connected across  $120\text{V}$ ,  $60\text{Hz}$  source.
- Compute the power factor of the load.
  - A capacitor is connected in parallel with the entire load so that no reactive power is delivered by the source. Compute the value of the capacitance.

1. a) AC voltage and current waveforms may be expressed mathematically as  $v = V_{max}\sin\omega t$  or  $i = I_{max}\sin\omega t$ , while the RMS (root mean square) value of a sinusoidal voltage or current may be expressed as

$$V_{rms} = \frac{V_{max}}{\sqrt{2}}.$$

Therefore, the rms value of  $v = 340\sin(628.318t + 0.5236)$ V is

$$V_{rms} = \frac{V_{max}}{\sqrt{2}} = \frac{340V}{\sqrt{2}} = 240.4163V$$

- b) Using the AC voltage/current formula given above, the angular frequency  $\omega$  is identified as the coefficient of the independent variable  $t$ .  $\omega$  is given in radians per second, and is connected to the frequency  $f$  in Hertz (cycles per second) by  $\omega = 2\pi f$ .

Therefore, the frequency of the given current is

$$\frac{\omega}{2\pi} = \frac{628.318}{2\pi} \approx 100\text{Hz}.$$

- c) The phase shift of a signal is incorporated into the basic formula for AC signals (using voltage as an example) as  $v = V_{max}\sin(\omega t \pm \theta)$ , where  $\theta$  refers to the signal's phase shift in radians. Positive phase angles shift the resultant waveform to the left along the time axis, while negative phase angles shift the resultant waveform to the right along the time axis.

For the given voltage, converting to degrees,  $\theta_V = 30^\circ$ , and for the given current,  $\theta_I = 50^\circ$ .

Therefore, the difference between the current and voltage phases is

$$\theta_{VI} = \theta_V - \theta_I = 30^\circ - 50^\circ = -20^\circ.$$

- d) The average value of a periodic signal, such as an AC voltage or current, may be found by averaging its value over a single period  $T$  of the signal. Averaging voltage over a single period  $T = \frac{1}{f} = \frac{1}{100\text{Hz}} = 10\text{ms}$ ,

$$V_{ave} = \frac{1}{T} \int_0^T A\sin(\omega t - \theta_v)dt.$$

Since the sine and cosine functions are symmetric over each period, this integral formula will always evaluate to 0 regardless of the values of  $A$ ,  $\omega$ , or  $\theta$ .

Therefore,

$$V_{ave} = 0V.$$

This is the reason rms values are more commonly used as a measure of the average value of an AC voltage or current.

- e) Load impedance  $Z$  is given by the sum of the real and imaginary components of impedance, resistance,  $R$ , and reactance,  $X$ ,

$$Z = R + jX,$$

and can be derived by expressing both voltage and current as phasors (assuming they have the same angular frequency  $\omega$  and using the Ohm's Law relationship

$$Z = \frac{V}{I} = \frac{340\angle 30^\circ \text{V}}{100\angle 50^\circ \text{A}} = 3.4\angle -20^\circ \Omega.$$

The negative phase angle of the impedance implies a capacitive load.

2. a) The power factor of a given load is given by the ratio between apparent and real power,

$$P.F. = \frac{P_{real}}{P_{apparent}},$$

where real and apparent power are given by

$$P_{apparent} = V_{rms}I_{rms} \quad (1)$$

$$P_{real} = V_{rms}I_{rms}\cos(\theta_{VI}). \quad (2)$$

To find the phase angle of the current, we may convert the complex impedance to phasor form,

$$Z = 4 - j2\Omega = \sqrt{20}\angle -26.5651^\circ$$

and use Ohm's Law in the form

$$I = \frac{V}{Z} = \frac{120\angle 0^\circ \text{V}}{\sqrt{20}\angle -26.5651^\circ \Omega} = 26.8328\angle 26.5651^\circ \text{A}.$$

Thus, the phase angle is

$$\theta_{VI} = -26.5651^\circ.$$

Finally, finding apparent power

$$V_{rms}I_{rms} = 120\text{V} * 26.8328\text{A} = 3.22\text{kVA}$$

and real power

$$P_{real} = P_{apparent}\cos(\theta_{VI}) = 3.22\text{kVA} * 0.8944 = 2.88\text{kW}.$$

Therefore, the power factor of the given load is

$$P.F. = \frac{P_{real}}{P_{apparent}} = \frac{2.88}{3.22} = 0.8944.$$

- b) Source current in phasor form was found in part (a). We cannot convert to the time domain since we are not given the angular frequency  $\omega$ ; thus, the source current is

$$I = 26.8328 \angle 26.5651^\circ \text{ A}$$

- c) Real power was found in part (a),

$$P_{real} = 2.88 \text{ kW}$$

- d) Reactive power  $P_{reactive}$  is given by

$$P_{reactive} = P_{apparent} \sin(\theta_{VI}) = 3.22 \text{ kVA} * -0.4472 = -1.44 \text{ kVAR}.$$

In addition, reactive power  $Q < 0$  implies a capacitive load.

3. a) Source frequency affects the reactance of both inductors and capacitors, defined as

$$X_L = 2\pi fL \quad (3)$$

$$X_C = \frac{1}{2\pi fC}. \quad (4)$$

Finding the reactance of the load inductor given,

$$X_L = 31.4034 \Omega.$$

Therefore, complex impedance in phasor form is given as

$$Z = 3.5 + j31.4034 \Omega = 31.5978 \angle 83.6405^\circ \Omega$$

and load current is found as

$$I = \frac{V}{Z} = \frac{120 \angle 30^\circ \text{ V}}{31.5978 \angle 83.6405^\circ \Omega} = 3.7977 \angle -53.6405^\circ \text{ A}$$

- b) Collecting needed values,

$$P_{apparent} = V_{\text{rms}} I_{\text{rms}} = 120 \text{ V} * 3.7977 \text{ A} = 455.7240 \text{ VA}$$

$$\theta_{VI} = \theta_V - \theta_I = 30^\circ + 53.6405^\circ = 83.6405^\circ$$

$$P_{real} = P_{apparent} \cos(\theta_{VI}) = 455.7240 \text{ VA} * 0.1108 = 50.4789 \text{ W},$$

and the load power factor is

$$P.F. = \frac{P_{real}}{P_{apparent}} = \frac{50.4789}{455.7240} = 0.1108.$$

- c) Real power was found in part (b),

$$P_{real} = 50.4789 \text{ W}.$$

d) Reactive power is

$$P_{reactive} = P_{apparent} \sin(\theta_{IV}) = 452.9196 \text{ VAR.}$$

4. a) In this case, with a source frequency of 50Hz, the reactance of the inductor will change to be

$$X_L = 26.1695 \Omega.$$

Recalculating the circuit's complex impedance and finding the load current,

$$I = \frac{V}{Z} = \frac{120 \angle 30^\circ \text{ V}}{26.4025 \angle 82.3823^\circ \Omega} = 4.5450 \angle -52.3823^\circ \text{ A.}$$

- b) The load power factor is then found by recalculating the real and apparent power and finding the ratio between them,

$$P.F. = \frac{P_{real}}{P_{apparent}} = \frac{72.2966 \text{ W}}{545.4 \text{ VA}} = 0.1326$$

- c) Real power is given in part (b) as

$$P_{real} = 79.2966 \text{ W.}$$

d) Reactive power is

$$P_{reactive} = P_{apparent} \sin(\theta_{IV}) = 540.5836 \text{ VAR.}$$

5. First, the total current in phasor form is found using the apparent power,  $R + jQ$ , as

$$I^* = \frac{R + jQ}{V} = \frac{3605.6 \angle 33.6901^\circ}{120 \angle 0^\circ} = 30.0467 \angle 33.6901^\circ \text{ A.}$$

This gives a total current, in phasor form, of  $I_{total} = 30.0467 \angle -33.6901^\circ \text{ A}$ . By Kirchhoff's current law, we can assume that  $I_{total} = I_1 + I_2$ , where  $I_1 = \frac{V}{Z_1} = \frac{120 \angle 0^\circ}{5 \angle 53.1301^\circ} = 24 \angle -53.1301^\circ \text{ A}$ . Therefore,

$$I_2 = I_{total} - I_1 = 10.6004 + j2.5331 = 10.8989 \angle 13.4395^\circ \text{ A.}$$

Then,

$$Z_2 = \frac{V}{I_2} = \frac{120 \angle 0^\circ}{10.8989 \angle 13.4395^\circ} = 11.0103 \angle -13.4395^\circ \Omega.$$

In rectangular coordinates,

$$Z_2 = 10.7088 - j2.5590 \Omega,$$

which implies that the second load is capacitive, though the overall load remains inductive.

6. a) First, the inductive reactance  $X_L$  is found as  $X_L = 2\pi fL = 0.3770\Omega$ , so the total impedance  $Z = 1 + j0.3770\Omega$ .

To find the power factor, we first find the phase angle difference between voltage and current,

$$\theta_{VI} = \text{atan}\left(\frac{0.377}{1}\right) = 20.6564^\circ.$$

Then, the power factor is found as

$$P.F. = \cos(\theta_{VI}) = 0.9357.$$

- b) Since the capacitor will be connected in parallel, it's most convenient to use susceptance,  $B$ , rather than reactance,  $X$ . To achieve a unity power factor  $P.F. = 1$ , the inductive and capacitive susceptances in parallel must be equal; that is,  $B_L = B_C$ . Finding the inductive susceptance,

$$B_L = \text{Im}(Y_L) = \text{Im}\left(\frac{1}{1 + j0.3770}\right) = 0.3301\text{S},$$

then setting the capacitive susceptance equal,

$$0.3301\text{S} = B_C = 2\pi fC \Rightarrow C = \frac{B_L}{2\pi f} = 875.62\mu\text{F}.$$